

NFL Prediction Model: Probabilistic Forecasting & Risk

Lecture 1: Forecasting as a Decision Problem

1.1 object of study is a distribution

Point forecast: Chiefs beat the bills

Distribution forecast: Chiefs margin $\sim N(\mu = 3, \sigma^2 = 13.5^2)$

$$P(\text{Chiefs win}) = \Phi(\mu/\sigma) = \Phi(3/13.5) = \Phi(.22) = 0.59$$

What if the sportsbook line is Chiefs -6.5 $\rightarrow$ they must win by 7 or more

$$P(\text{Chiefs cover the spread}) = 1 - \Phi\left(\frac{6.5 - 3}{13.5}\right) = 0.40$$

$\rightarrow$ Yields probability of various events (e.g. win, cover, etc.)

* A distribution is a reusable asset

1.2 The Forecasting Pipeline

Measure $\rightarrow$ Predict $\rightarrow$ Evaluate $\rightarrow$ Decide $\rightarrow$ Stress $\rightarrow$ Monitor

* Design Principle: Keep the forecast separate from the decision

1.2.1

• The output at each stage is a different kind of object

◦ "Measure": estimates of hidden quantities

◦ "Predict": probabilities of observed events

◦ "Evaluate" only looks at the forecast

◦ were the probabilities honest?

• The chain loops

◦ monitor feeds back into measure

1.2.2

• A decision problem has 2 ingredients:

1. **Beliefs**: a probability distribution p over what might happen

2. **Consequences**: a loss function, $L(a, \text{outcome}) \rightarrow$ result of a under each outcome

$$a^* = \arg \min_{a \in A} E_{Y \sim p}[L(a, Y)]$$

- a^* : optimal action

- A : set of available actions

- Y : uncertain outcome, random variable

- p : forecast distribution of Y

- $L(a, Y)$: loss from action a when outcome is y

- $E_{Y \sim p}[\cdot]$: expected value of r.v. Y

1.3 Two Kinds of Uncertainty

1) **Alentory Uncertainty**: irreducible randomness in the world

- tipped passes & fumbles

- no model removes this type of uncertainty

2) **Epistemic Uncertainty**: uncertainty about the model itself

- true μ is 1.1 rather than 1.4

- reduced with more data & better methods

1.4 Signal, Noise, NFL

- NFL: low signal-to-noise ratio

- short seasons, low score margin, roster change

- a good model only picks 63-66% of winners

- Consequences

- 1) Judge the process, not individual outcomes

- 2) Beware results that look too good

- 3) Humility is part of the method